Project: Worked examples
Title: Thin cylindrical shell under internal pressure
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the minimum thickness of a thin cylindrical shell under internal pressure, for circumferential and longitudinal stress. It adds corrosion allowance and a design factor, and compares the thickness provided with the requirement.

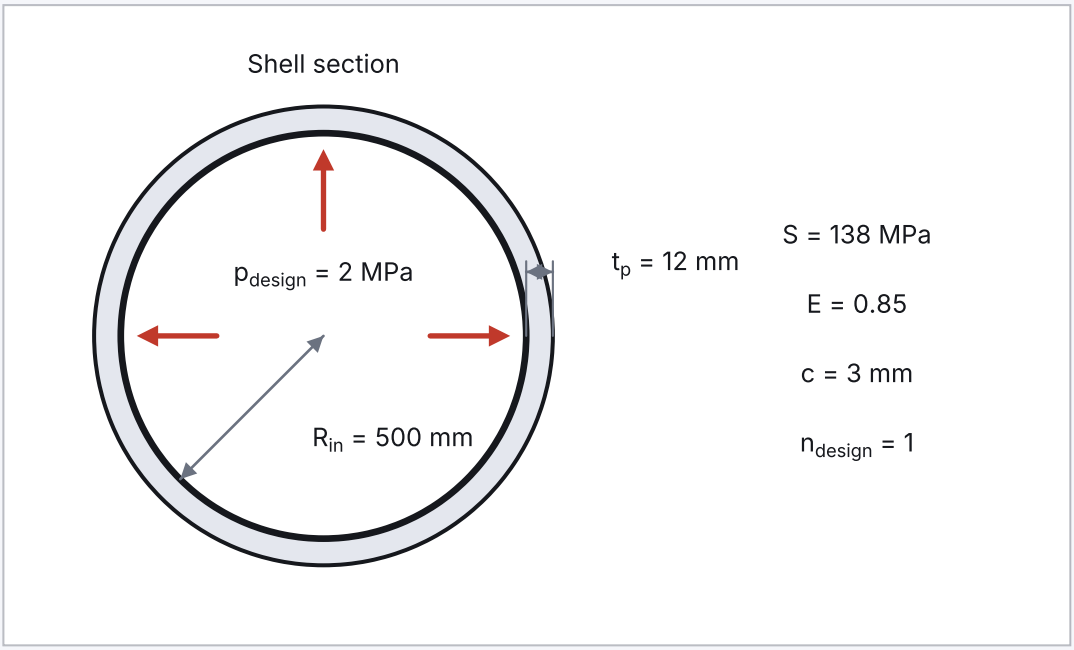


Figure 1 — A thin-walled cylinder in section under internal pressure: inside radius R_{in}, shell thickness t_p including the corrosion allowance c. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Thickness needed for the circumferential stress	$t_c = p_{design} \cdot R_{in} / (S \cdot E_j - 0.6 \cdot p_{design})$
Thickness needed for the longitudinal stress	$t_l = p_{design} \cdot R_{in} / (2 \cdot S \cdot E_j + 0.4 \cdot p_{design})$
Required thickness, with the design factor and corrosion allowance	$t_{req} = n_{design} \cdot \max(t_c, t_l) + c$
Hoop stress in the plate provided, after corrosion	$\sigma_\theta = p_{design} \cdot (R_{in} + 0.6 \cdot t) / (E_j \cdot t)$, at $t = t_p - c$
Range of validity of the thin-wall formulae	$t_c \le R_{in}/2$ and $p_{design} \le 0.385 \cdot S \cdot E_j$

Basis: Thickness of shells under internal pressure, circumferential stress[1](cl. UG-27(c)(1)); Thickness of shells under internal pressure, longitudinal stress[1](cl. UG-27(c)(2)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
R_{in}	Inside radius (corroded)	500	mm	drawing: DRG-PV-601 rev A	5 … 10000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
t_p	Shell thickness provided (nominal, before corrosion)	12	mm	drawing: DRG-PV-601 rev A	0.5 … 500 mm
S	Allowable stress at design temperature	138	MPa	spec: SPEC-MECH-03 design criteria	10 … 1000 MPa
p_{design}	Internal design pressure	2	MPa	spec: LOAD-SCHEDULE-01	0.01 … 100 MPa
c	Corrosion allowance	3	mm	spec: SPEC-MECH-03 design criteria	0 … 25 mm
E_j	Joint efficiency	0.85		spec: SPEC-MECH-03 design criteria	0.45 … 1
n_{design}	Design factor on the computed thickness (legacy; 1 for none)	1		spec: SPEC-MECH-03 design criteria	1 … 5

3. Assumptions

A1. The shell is cylindrical, away from heads, nozzles and supports, and carries membrane stresses only. (The scope of the UG-27 formulae.)
A2. The joint efficiency reduces the allowable stress uniformly. (Code-form simplification carried from the legacy module.)

4. Working

[8] $t_c = p_{design} * R_{in} / (S * E_j - 0.6 * p_{design})$ [1](VIII-1 UG-27(c)(1))
= 2 MPa × 500 mm / (138 MPa × 0.85 − 0.6 × 2 MPa)
= 8.6133 mm
Thickness for circumferential stress

[9] $t_l = p_{design} * R_{in} / (2 * S * E_j + 0.4 * p_{design})$ [1](VIII-1 UG-27(c)(2))
= 2 MPa × 500 mm / (2 × 138 MPa × 0.85 + 0.4 × 2 MPa)
= 4.2481 mm
Thickness for longitudinal stress

[10] $t_{req} = n_{design} * \max(t_c, t_l) + c$
= 1 × max(8.6133 mm, 4.2481 mm) + 3 mm
= 11.613 mm
Required nominal thickness (design factor and corrosion allowance included)

[11] $t_{corr} = t_p - c$
= 12 mm − 3 mm
= 9 mm
Corroded thickness provided

[12] $\sigma_h = p_{design} * (R_{in} + 0.6 * t_{corr}) / (E_j * t_{corr})$ [1](VIII-1 UG-27(c)(1))
= 2 MPa × (500 mm + 0.6 × 9 mm) / (0.85 × 9 mm)
= 132.13 MPa
Hoop stress at the corroded thickness provided

[13] $u = t_{req} / t_p$
= 11.613 mm / 12 mm
= 0.96777
Required over provided thickness

5. Results

SYMBOL	VALUE	DESCRIPTION
t_{req}	11.61 mm	Required nominal thickness (governing)
σ_h	132.1 MPa	Hoop stress at the corroded thickness
t_c	8.613 mm	Thickness, circumferential stress
t_l	4.248 mm	Thickness, longitudinal stress
u	0.97	Thickness utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Provided thickness against required	11.61 mm	12 mm	0.97	[1](VIII-1 UG-27)

7. Applicability

None.

8. Verification

Template mech.thin-vessel@1.1.0 Content hash 4e7be0d562d46f01 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Boiler and Pressure Vessel Code, Section VIII, Division 1: Rules for Construction of Pressure Vessels. ASME.